

Perceptually-Based Optimization for Radiometric Projector Compensation

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ABSTRACT

Radiometric compensation techniques have been proposed to manipulate the appearance of arbitrarily textured surfaces using projectors. However, due to the limited dynamic range of the projectors, these compensation techniques often fail under bright environmental lighting or when the projection surface contains high contrast textures, resulting in clipping artifacts. To address this issue, we propose to apply a perceptually-based tone mapping technique to generate compensated projection images. The experimental results demonstrated that our approach minimizes the clipping artifacts and contrast degradation under challenging conditions.

Index Terms: Computing methodologies [Computer graphics]: Graphics systems and interfaces—Perception; Computing methodologies [Computer graphics]: Graphics systems and interfaces—Mixed / augmented reality

1 INTRODUCTION

Projection mapping is a promising technique that can edit the appearance of real objects' surfaces using projectors [8]. To manipulate the appearance of arbitrarily textured surfaces, radiometric compensation techniques have also been proposed [2]. However, due to the limited dynamic range of the projectors, these compensation techniques often fail under bright environmental lighting or when the projection surface contains high contrast textures, resulting in clipping artifacts. To overcome this problem, several previous works have proposed adaptive compensation techniques that compress the intensities of an input image in a spatially adaptive manner such that the clipping artifacts become less visible [1, 3]. One drawback of these techniques is that the contrast may be too small in the projection results because the input image is simply scaled to fit into the dynamic range. Also, these algorithms do not have legitimate criterion as to which should be prioritized between contrast preservation and clipping artifact reduction when clipping artifacts are unavoidable.

To tackle the above challenges, we reframe the adaptive radiometric compensation problem as the tone mapping from high to low dynamic range and apply a perceptual optimization algorithm to compute the projected image. Specifically, we optimize the projected image so that the target appearance and the simulated projection result are closest in a perceptual distance model, named Normalized Laplacian Pyramid Distance (NLPD) [7]. This model simulates the low-level human visual processing such as the band-pass decomposition and nonlinear gain control and represents supra-threshold contrast perception. Therefore, as a result of optimization, the contrast of projected images can be locally enhanced at each spatial frequency to be perceptually closest to the target within a given

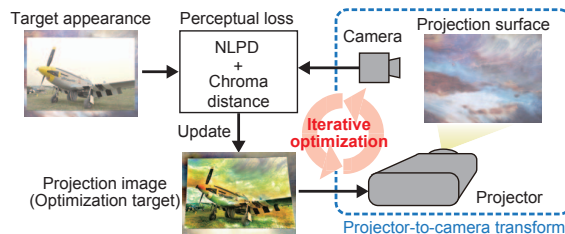


Figure 1: Overview of the proposed method.

dynamic range. In addition, the balance between the visibility of the clipping artifacts and the contrast of the target image is automatically resolved by considering the distance to the target image in the perceptual representation. While typical tone mapping problems only consider conversions between globally different dynamic ranges, we show that the proposed method is effective even when the dynamic range is locally different due to the spatially varying reflective properties of the projection surface.

2 METHOD

Figure 1 presents the overview of the proposed method. We assume that the intensity of both the projector and camera are linearized. Given the target appearance $C_{tgt} \in \mathbb{R}^{H \times W \times 3}$ in the camera perspective, our method optimizes the projection image $P \in \mathbb{R}^{H \times W \times 3}$ so that it gives the perceptually most similar appearance to C_{tgt} after projection onto the given surface as:

$$P_{opt} = \underset{P}{\operatorname{argmin}} D(C_{tgt}, f(P)), \quad (1)$$

where $D(\cdot, \cdot)$ is a perceptual distance between the two images and f is a function that simulates the appearance after projection given a projected image P . We use NLPD [7] for the perceptual distance function. However, since NLPD only considers luminance information, we additionally incorporate the Euclidean distance in chroma channels of CIE $L^*a^*b^*$ color space, $\Delta E_{a^*b^*}$, to account for perceived color differences. Taken together, our perceptual distance function can be expressed as $D(A, B) = NLPD(A, B) + \Delta E_{a^*b^*}(A, B)$.

To simulate the appearance of the projection result, the function f first geometrically transform P into the camera perspective image P' using a projector-to-camera mapping function obtained by the structured light technique. Then, the colors of the image in each pixel is converted by multiplying the color-mixing matrix, $V \in \mathbb{R}^{3 \times 3}$, and the diagonal reflectance matrix, $R \in \mathbb{R}^{3 \times 3}$, and then adding the radiance from the surface due to the ambient light, $E \in \mathbb{R}^{3 \times 1}$ as $RV P' + E$. R , V , and E are estimated using the conventional calibration procedure such as [2]. Although we didn't take into account projector defocus, one may additionally incorporate this as in [11] when projecting on a surface with large depth variations. One can also model the entire projector-camera transform using a light transport matrix [10] for improved precision.

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Figure 2: Comparison of projection results. Best viewed in zoom.

3 EXPERIMENT

Experimental setup The apparatus we used consisted of a laptop computer (MSI WS-66, equipped with NVIDIA Quadro RTX5000 Max-Q design 16GB GPU), a camera (Point Grey Research Flea3, 1600×1200 pixels), and a projector (EPSON EH-TW5650, 1280×720 pixels). We set the dynamic range of the projector on a white reference to the range of $117\text{--}876\text{ cd/m}^2$ assuming a challenging scenario where perfect compensation is impossible. We arranged the camera-projector configuration so that every single pixel of the projector was covered by multiple pixels of the camera. In this way, the optimized projection image exhibits the image sharpening effect that compensates for the reduced resolution. A target image C_{tgt} was generated by editing the reflectance of the projection surface. Specifically, we first replaced a portion of area of the reflectance K with a digital image taken from Kodak Image Dataset. Then, we simulated its appearance under the maximum light projection to obtain C_{tgt} .

Optimization The projection image P was iteratively optimized using Adam optimizer [6] implemented in PyTorch. We set the learning rate to 0.02 and left the other parameters to the default values: $(\beta_1, \beta_2) = (0.9, 0.999)$. The projection image was initialized by a uniform random noise between $[0.3, 0.7]$. For each update, each intensity value was clipped in the range of $[0, 1]$ and quantized to 8-bit precision in the gamma-encoded space. The optimization was finished after 1000 iterations. The optimization of a projection image with 1280×720 pixels took 72 secs.

Results In Figure 2, we compared the results of our proposed method (NLPD-optimized) to three different baselines. In *Standard compensation*, the projected images were obtained by simply inverting the projector-to-camera transform function (i.e., $P = f^{-1}(C_{\text{tgt}})$) and then clipping the intensity range into $[0, 1]$. In *Adaptive compensation*, the intensity range of the target image is scaled using the algorithm of [1]. In *SSIM-optimized*, we replaced the perceptual distance function defined by NLPD with that by SSIM [9], which has been used for objective evaluation of compensated results as well as the loss function to train neural networks that generate a compensated projection image [4]. As SSIM takes the highest value of 1 at the perfect similarity, we used $1 - \text{SSIM}$ for the distance function. The results show that the proposed approach was the most successful in compensating for the surface textures while maintaining as much contrast in the target appearance as possible.

4 APPLICATION TO DEFORMATION EFFECT

We also applied the proposed method to deform a printed material without changing the original appearance, similarly to [5]. The deformation effects are one of the most challenging scenarios for projection AR because displacing high contrast patterns requires a high

dynamic range projection. In the supplementary video, we show that our perceptual optimization can improve the motion impressions.

5 DISCUSSION AND FUTURE WORK

The main drawback of the proposed method is the long processing time (a few minutes at full HD resolution) required for optimization of a projected image. One possible solution to this problem is to train a feedforward network that can approximate the iterative optimization process in the calibration step, as done by [4]. Incorporating the temporal aspect of the HVS into the perceptual model will also be important when dealing with video inputs. Finally, the color fidelity of the projection results may be further improved by incorporating a more sophisticated color appearance model.

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